Aaryan Lath

Systems Engineer | aaryanlath.com | linkedin.com/in/aaryan-lath | (312) 838-5669 | aaryanlath05@gmail.com

EDUCATION

Purdue University — **West Lafayette, IN**

Bachelor of Science in Aeronautical and Astronautical Engineering *(GPA: 3.74/4.0)* — *Expected May 2026*

Master of Science in Aeronautics and Astronautics *(GPA: 4.0/4.0)* — *Expected May 2027*

Specialization in Aerodynamics and Systems Design

Honors: Dean's List and Semester Honors for **every** semester, AAE Ambassador

SKILLS

Core Tools: Technical Writing, CAD, Assembly, CREO, NX, GD&T, Teamcenter, Ansys, Manufacturing, JIRA
Programming & Analysis: System Integration, Python, MATLAB, Git, CFD, Simulink, LabView
Soft-skills: Analytical, Communicator, Detail-oriented, Fast-paced learner, Inquisitive
Awards: 2nd place in Nationals of World Robot Olympiad Junior High

PROFESSIONAL EXPERIENCE

Siemens USA — **Grand Prairie, TX**

Systems Engineering Intern — *May 2025 – Aug 2025*

- Designed custom enclosures for panelboards using **CREO** and executed ECNs in **SAP**.
- Streamlined **100+** switchboard neutral assembly configurations by resolving edge case designs.
- Automated BOM generation with Python, cutting manual processing time by **~30%** for **10+** orders/day.

Purdue School of Aeronautics and Astronautics — **West Lafayette, IN**

Undergraduate Teaching Assistant — *Jan 2025 – Present*

- Led study sessions to teach **50+** students, core **AAE 251** (Aircraft and Spacecraft Design) course material.
- Provided personalized support to students with concepts and **MATLAB** troubleshooting.
- Guided students in understanding key design principles for the aircraft and spacecraft design project.

Resilient Extraterrestrial Habitat Institute — **West Lafayette, IN**

Undergraduate Research Assistant - Systems Engineer — *May 2024 – May 2025*

- Analyzed **18** potential disruptions scenarios in a lunar habitat through **Simulink** simulations.
- Performed trade studies to evaluate safety controls, optimizing **resilience** and costs using **ESM** analysis.
- Executed **FEA** simulations and tests to validate **vibration isolation** devices under various conditions.

PROJECTS AND INVOLVEMENTS

SAE Aero Design — **West Lafayette, IN**

Chief Engineer / Project Manager — *May 2025 – Present*

- Documented past failures from the SAE Aero Design Regular Class to avoid recurring errors.
- Authored a **75-page** RC aircraft design guide to accelerate onboarding and designing an onboarding project.
- Led cross-functional development for **3** teams by reviewing trade studies, ensuring milestones/goals are met.
- Bringing-in **3+** sponsors into the club along with creating, teaching the business team to expand the club.

Purdue Space Program – A SEDS Chapter — **West Lafayette, IN**

Technical Director — *Dec 2024 – May 2025*

- Optimized the organization's backend operations and managed **9** technical teams.

Satellites Deputy Systems Director — *Feb 2024 – Nov 2024*

- Improved project progress with **~15** engineers and other sub-teams ensuring alignment with project goals.
- Formulated, reviewed, and refined **200** project and system-level requirements for Boiler Bus.
- **Manufactured** the satellite frame end plates using **CAM** and walls through **laser cutting**.

Purdue Aerial Robotics Team — **West Lafayette, IN**

Competition Airframe — *Oct 2023 – May 2024*

- Engineered aircraft parts for the AUVSI SUAS competition through **NX** and Teamcenter with **~30** engineers.
- Applied engineering solutions to resolve manufacturing issues of molds, carbon fiber, and internal structures.

RESEARCH PAPERS

Coauthoring "A Modular MDO Framework for Rapid Electric Aircraft Design". Awaiting publishing approval as a conference paper for the AIAA Aviation 2026

"A Systems-Theoretic Safety Analysis of DEP Architectures in High-Density Airspace.", conference style paper for AAE 571 - Complex Systems Safety